

EE3621: Digital Signal Processing

Lecture 24: Design of FIR filter - frequency sampling (III B.Tech EEE)

September 23, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Sample** desired frequency responses $H_d(e^{j\omega})$ with proper linear-phase phase assignments.
 2. **Compute** FIR filter coefficients $h[n]$ via IDFT formulation.
 3. **Optimize** transition band samples to maximize stopband attenuation.
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2. MATHEMATICAL FOUNDATIONS

2.1 Frequency-Sampling Design Equations

Let $H[k] = H_d(e^{j\frac{2\pi k}{N}})$. For a linear-phase FIR filter of length N :

$$H[k] = |H[k]|e^{j\angle H[k]}, \quad \angle H[k] = -\left(\frac{N-1}{2}\right)\frac{2\pi k}{N}$$

The impulse response is computed by the IDFT:

$$h[n] = \frac{1}{N} \sum_{k=0}^{N-1} H[k]e^{j\frac{2\pi kn}{N}}$$

Exploiting Hermitian symmetry ($H[N-k] = H^*[k]$):

- **For N Odd:**

$$h[n] = \frac{H[0]}{N} + \frac{2}{N} \sum_{k=1}^{(N-1)/2} |H[k]| \cos \left[\frac{2\pi k}{N} \left(n - \frac{N-1}{2} \right) \right]$$

- **For N Even:**

$$h[n] = \frac{H[0]}{N} + \frac{2}{N} \sum_{k=1}^{N/2-1} |H[k]| \cos \left[\frac{2\pi k}{N} \left(n - \frac{N-1}{2} \right) \right] + \frac{H[N/2]}{N} (-1)^n$$

2.2 Transition Sample Optimization Values

Number of Transition Samples	Optimized Sample Values	Resulting Stopband Attenuation
0 (No transition band)	None ($H[k] \in \{0, 1\}$)	-16 dB
1 Transition Sample	$T_1 \approx 0.38$	-45 dB
2 Transition Samples	$T_1 \approx 0.59$; $T_2 \approx 0.11$	-75 dB
3 Transition Samples	$T_1 \approx 0.70$; $T_2 \approx 0.25$; $T_3 \approx 0.02$	-95 dB

Pictorial Representations

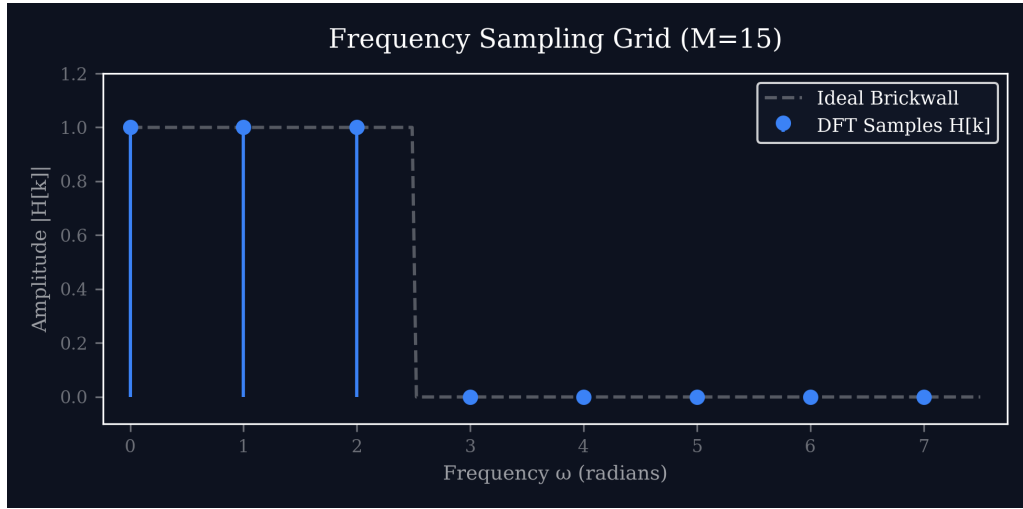


Figure 1: Freq Sampling Discrete

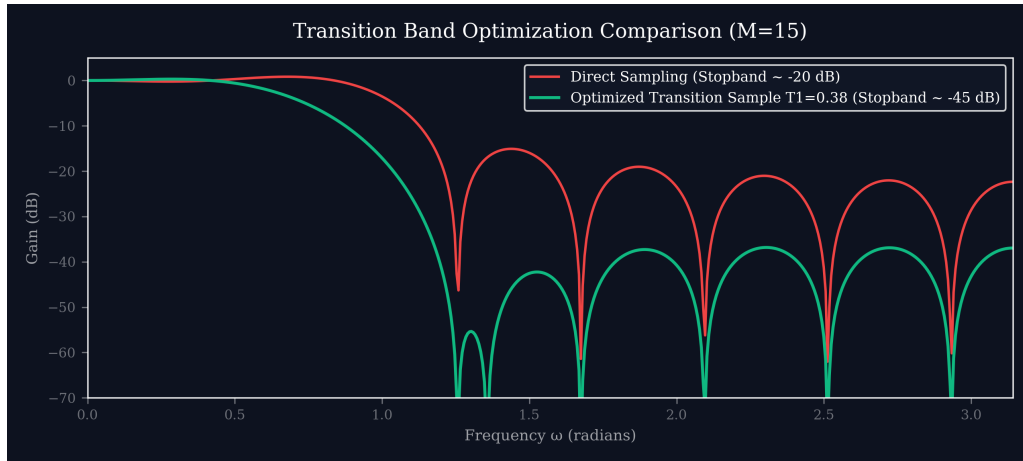


Figure 2: Transition Samples

3. WORKED NUMERICAL EXAMPLES

Example 24.1: Frequency-Sampling Lowpass Filter Design

Problem: Design a 7-point ($N = 7$) linear-phase FIR lowpass filter with frequency samples:

$$|H[k]| = 1, 1, 0, 0, 0, 0, 1 \quad \text{for } k = 0, 1, 2, 3, 4, 5, 6$$

Solution:

- Length $N = 7$ (Odd), $\tau = \frac{7-1}{2} = 3$.
- Non-zero samples: $|H[0]| = 1$; $|H[1]| = |H[6]| = 1$.
- Using the real summation formula:

$$h[n] = \frac{H[0]}{7} + \frac{2}{7}|H[1]| \cos \left[\frac{2\pi(1)}{7}(n-3) \right] = \frac{1}{7} + \frac{2}{7} \cos \left[\frac{2\pi}{7}(n-3) \right]$$

$$- h[3] = \frac{1}{7} + \frac{2}{7} \cos(0) = \frac{3}{7} \approx 0.4286$$

$$- h[2] = h[4] = \frac{1}{7} + \frac{2}{7} \cos \left(\frac{2\pi}{7} \right) = \frac{1+2(0.6235)}{7} = \frac{2.2470}{7} \approx 0.3210$$

$$- h[1] = h[5] = \frac{1}{7} + \frac{2}{7} \cos \left(\frac{4\pi}{7} \right) = \frac{1+2(-0.2225)}{7} = \frac{0.5550}{7} \approx 0.0793$$

$$- h[0] = h[6] = \frac{1}{7} + \frac{2}{7} \cos \left(\frac{6\pi}{7} \right) = \frac{1+2(-0.9010)}{7} = \frac{-0.8019}{7} \approx -0.1146$$

$$h[n] = -0.1146, 0.0793, 0.3210, 0.4286, 0.3210, 0.0793, -0.1146$$

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